

Saqib Ali Shah

Software Engineer

+92 347-5287822 — saqibshah2882@gmail.com — saqib-002.github.io — linkedin.com/in/saqibshah172 — github.com/Saqib-002

Summary

Software Engineer with a proven track record of architecting scalable full-stack systems and high-performance Agentic RAG workflows. Specialized in engineering intelligent infrastructure, including B2B lead platforms and secure natural language database interfaces using Python, FastAPI, React, PostgreSQL, and Gemini Vertex AI on GCP and AWS. Experienced with LangChain, LangGraph, LlamaIndex, Docker, and vector databases (Pinecone, Weaviate). Focused on transforming complex data bottlenecks into production-ready, AI-driven products that deliver measurable cost savings and sub-second performance.

Education

Institute of Space & Technology <i>Bachelor of Science in Computer Science</i> • Gold Medalist: Awarded for securing 1st position in the graduating batch for academic excellence.	Islamabad, Pakistan Sep 2020 – Jul 2024
KRL Model College For Boys <i>HSSC (Pre-engineering)</i>	Kahuta, Pakistan 2018 – 2020

Experience

Software Engineer – Freelance <i>Independent Contractor (Remote)</i> • Modernized retail and F&B business operations by deploying custom POS and e-commerce solutions, streamlining order workflows and inventory management. • Automated document processing pipelines to replace manual data entry, reducing administrative processing time by an estimated 60% and eliminating manual errors. • Fine-tuned AI vision models for intelligent invoice parsing and financial record extraction, achieving over 90% data extraction accuracy and reducing processing time by 3x compared to manual entry. • Built intuitive interfaces enabling non-technical staff to manage high-volume transactions and digital storefronts effortlessly.	Oct 2025 – Present
Software Engineer <i>Modern Intelligent Solutions</i> • Architected Imagini, a high-performance B2B leads platform housing millions of leads records; engineered a credit-based access system and streaming upload pipelines using React and FastAPI to maintain sub-second search and filtering at scale. • Developed a multi-agent RAG framework to navigate 1.5M Arabic academic documents, implementing a hierarchical meta-agent with graph-based traversal and Arabic-tuned embeddings to handle sophisticated linguistic queries with high precision. • Implemented a secure natural language database interface using LangGraph and Gemini Vertex AI, implementing strict intent classification and multi-layer guardrails to prevent unauthorized data modification. • Designed an RFM-based customer segmentation engine for a premium airport services firm, deploying custom predictive scoring models to automate dynamic subscription plans and reduce operational costs. • Built a large-scale cold outreach pipeline processing ~1B raw leads on GCP; automated personalized email and SMS generation using Gemini Vertex AI with a TypeScript web crawler for real-time lead enrichment.	Jul 2024 – Dec 2025 Riyadh, Saudi Arabia
ML Engineer <i>Voilabs</i> • Fine-tuned transformer models for OCR-free document data extraction and optimized the architecture for mobile deployment. • Integrated ML models into web applications via reusable components, improving AI feature delivery speed across client projects. • Built data processing pipelines and fine-tuned neural networks to improve task-specific model accuracy by 25%.	Nov 2023 – Feb 2024 France (Freelance)
Software Engineer Intern <i>Ksoft</i> • Implemented real-time object detection solutions using computer vision models. • Designed intuitive user interfaces and contributed to mobile application development to read & parse business cards.	Aug 2023 – Sep 2023 Islamabad, Pakistan

Technical Projects

ThunderJaw Restaurant POS — <i>Electron, React, TypeScript, PostgreSQL, Knex</i> • Developed a cross-platform desktop application for restaurant management featuring real-time order tracking, kitchen management, and automated PostgreSQL database migrations. • Implemented secure role-based access control (RBAC) and modular state management to handle complex inventory, worker salaries, and delivery logistics in high-traffic environments.
High-Scale Email Verification System — <i>Python, GCP, Docker, Celery, FastAPI</i> • Engineered a multi-layered validation pipeline (DNS resolution, SMTP probing, and heuristics) that reduced verification costs by 90% compared to commercial providers. • Implemented asynchronous workers and distributed task queues to process hundreds of thousands of emails concurrently while avoiding IP blacklisting.
Mandumah Multi-Doc Agent — <i>LlamaIndex, Gemini Vertex AI, FastAPI, Python</i>

- Developed an agentic RAG system to navigate 1.5M Arabic academic documents using a hierarchical meta-agent architecture and graph-based traversal.
- Optimized retrieval for complex Arabic linguistic nuances, leveraging tool-calling for automated document summarization and deep-dive query answering.

B2B Leads Platform (Imagini) — *React, FastAPI, PostgreSQL, Redis, Celery*

- Architected a scalable platform managing millions of leads, featuring streaming data ingestion and a credit-based access system for B2B users.
- Optimized database indexing and caching to maintain sub-second search and filtering performance across massive datasets.

Certifications

Generative AI Fundamentals – *Google*

AI For Everyone – *Coursera*

Pandas – *Kaggle*

Applied Software Engineering Fundamentals – *Coursera*

Linear Algebra for Machine Learning and Data Science – *Coursera*

Calculus for Machine Learning and Data Science – *Coursera*

Backend Development & DevOps – *Boot.dev*

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3, Go

Frontend & Desktop: React.js, React Native, Next.js, Redux, TailwindCSS, Material UI, Electron

Backend & Cloud: Node.js, FastAPI, Express.js, Docker, Flask, AWS, Google Cloud Platform (GCP)

AI & Machine Learning: PyTorch, LangChain, LangGraph, LlamaIndex, Gemini Vertex AI, RAG Pipelines

Databases & Tools: MongoDB, PostgreSQL, Redis, Supabase, Pinecone, Weaviate, Celery, Knex.js (ORM)

Concepts & Methods: Retrieval-Augmented Generation (RAG), Multi-Agent Systems, REST APIs, Microservices, CI/CD, RBAC, NLP, Computer Vision, OCR, Prompt Engineering, RFM Segmentation, Distributed Systems